



Physics H Review Guide

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Range: Chapter 18-20.

Focus: 19.2(2)Combined circuit + 20.1-20.7 (20.4 partA only)

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Chapter 18 – Electric Currents

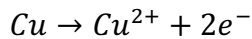
18.1 The Electric Battery

History

- 1780s, **Luigi Galvani** carried out a series of experiments on the contraction of a frog's leg muscles through electricity produced by static electricity
- Galvani conclusion: the source of electric charge was in the frog muscle/nerve itself and that the metal merely transmitted the charge to the proper points
 - Named this charge "animal electricity"
- **Alessandro Volta** believed: the source of electricity was rather in the contact between the dissimilar metals, or, a result of chemical reaction (oxidation + reduction)
- Experiment: Placed a piece of cloth soaked in salt solution or dilute acid between a disc of zinc and one of silver, and piled a "battery" of such couplings
 - When strips of metal connected to the two ends of the pile were brought close, a spark was produced
 - So, designed and built the first electric battery; published discovery in 1900

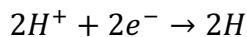
Electric Cells and Batteries

- Battery: produces electricity by transforming chemical energy to electrical energy; (several electric cells connected)
- Electric Cell Structure:
 - Two electrodes: plates or rods made of dissimilar metals
 - ✧ Positive (+) Cathode: e.g. Carbon, Copper, Silver
 - ✧ Negative (-) Anode: e.g. Zinc, Aluminum
 - Electrolyte: immersed solution (wet batteries) or paste (dry batteries)
 - ✧ E.g. Sulfuric acid (WET)
 - ✧ Usually use dry cells because wet cells are easy to spill or leak, which is not safe and not convenient
 - Terminal: the part of each electrode outside the solution
- How does an electric cell work?
(*In this example, I will use sulfuric acid for electrolyte, zinc for anode, and copper for cathode*)
 1. Acid dissolves the anode:
$$H_2SO_4 \rightarrow 2H^+ + SO_4^{2-}$$
$$Zn \rightarrow Zn^{2+} + 2e^-$$
 2. Two electrons stay behind, making the anode negative;
Positive ion Zn^{2+} enters H_2SO_4 , making the acid positive:
$$Zn^{2+} + SO_4^{2-} \rightarrow ZnSO_4$$
$$\therefore \text{Acid} = ZnSO_4 \text{ \& } 2H^+$$
 3. Electrons are pulled off the carbon cathode:



4. Positive ion Cu^{2+} stays behind, making the cathode positive;

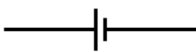
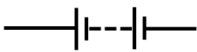
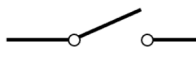
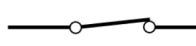
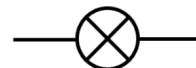
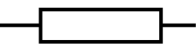
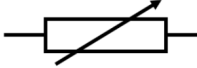
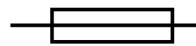
Two electrons enter acid, making the acid neutral again:



- The opposite charges on the two electrodes create a potential difference, and so CE is made available as EE
 - ✧ If terminals not connected: only a small amount of zinc is dissolved (since as the zinc anode becomes increasingly negative, new positive zinc ions produced will be attracted back to the anode), thus, a voltage is maintained
 - ✧ If terminals are connected: more zinc can be dissolved (since charge is allowed to flow between terminals), thus, one electrode will be used up after time and the cell becomes “dead”
- The voltage that exists depends on material of electrode and their relative ability to be dissolved or give up electrons
- Smaller cells contain fewer reactants and produce less EE, but voltage doesn't change
- When two or more cells are connected with positive terminal of one to negative terminal of another, they are connected in series, and voltages add up
 - $EMF_{\text{battery}} = \text{num}(\text{cells}) \times EMF_{\text{cell}}$

18.2 Electric Current

Circuit Symbols

				
Cell	Battery	Open Switch	Closed Switch	Filament Bulb
				
Fixed Resistor	Variable Resistor	Voltmeter	Ammeter	Fuse

Electric Circuit

- Electric Circuit: A continuous conducting path connected between terminals of a battery (so, with potential difference)
 - Elements: battery, device ran by EE (resistance), conducting wires, switch
- Three types of circuits:
 - Complete circuit: A continuous conducting path
 - Open circuit: There is a break in the circuit (No current flow)
 - Short circuit: Two wires (that should not have touched) touched (current shortened)
- Ground: In many real circuits, wire are connected to a common conductor to provide continuity

Electric Current

- Purpose of battery is to produce potential difference, which can make charges move
 - Potential difference = voltage, so voltage provides the force that pushes charge through a circuit
- When a complete circuit is formed, any flow of charge through the wires is called an electric current
- Electric current: The net amount of charge that passes through the wire's full cross section at any point per unit time
 - $I = \frac{\Delta Q}{\Delta t}$
 - I : Electric current (A)
 - ΔQ : Amount of charge that passes through (C)
 - Δt : Time interval (s)
 - Unit: Ampere (A), after **Andre Ampere**
 - ✧ $1 \text{ A} = 1 \text{ C/s}$
 - ✧ $1 \text{ mA} = 10^{-3} \text{ A}$; $1 \mu\text{A} = 10^{-6} \text{ A}$
 - SCALAR
- Direction of electric current is constant
 - Conventional Current: The direction positive charge would flow (High potential to low potential)
 - ✧ Was determined when electrons weren't discovered and so today, we still use the historical convention of positive charge flowing (but positive charges flowing in one direction is exactly equivalent to negative charges flowing in opposite direction)
 - Electron current: The direction of electron flow (Low potential to high potential)
- Conservation of electric charge: Charge doesn't disappear; a steady current at any instant is the same at one point as at any other point
 - A battery does not create (or destroy) any charge, a lightbulb does not destroy (or create) any charge
- Ammeter: device that measures electric current
 - Must be connected in series
 - $R_{\text{ammeter}} \approx 0$, so could be seen as a wire
 - Measurement Range: 0-3A and 0-0.6A (Use larger range for prevention of damage first, then change to smaller range)
- Galvanometer: measures smaller current

18.3 Ohm's Law: Resistance and Resistor

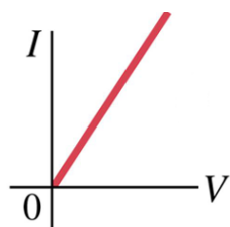
Electric Potential Difference/Voltage

- Higher voltage, more energy to push electrons out (greater flow of current)

- $E = QV$
 - E : Energy (J)
 - Q : Charge (C)
 - V : Potential Difference/Voltage (V)
- Voltmeter: Device that measures potential differences
 - Connected in parallel
 - Measurement Ranges: 0-15V and 0-3V

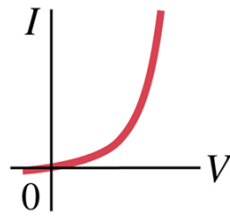
Ohm's Law

- **Georg Simon Ohm** established that: the current in a metal wire is directly proportional to the potential difference applied to its two ends
 - $I \propto V$
- Ohm's Law:
 - $V = IR$
 - V : Voltage (V)
 - I : Current (A)
 - R : Electrical Resistance (Ω)
- Electrical Resistance: The ratio of potential difference across a resistor to the current
- Resistors: Used to control the amount of current
 - Range: $<1\Omega \sim >10^6\Omega$
 - Main types:
 - ✧ "Wire-wound" resistors: consist of a coil of fine wire
 - ✧ "Composition" resistors: usually made of carbon and thin carbon or metal films
- Ohm's Law is not a fundamental law, but rather a description of a certain class of materials: metal conductors
 - Precondition: R should be a constant independent of V
 - Materials that don't have a constant R are said to be nonohmic, including semiconductor diodes and filament lamps
 - Given an IV graph, the slope would be $\frac{1}{R}$
 - ✧ Conductors' R is constant



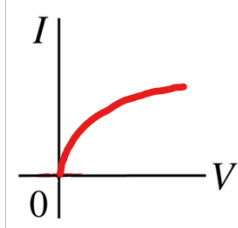
Thus, straight line for IV graph (linear relationship)

- ✧ Semiconductor diodes' R decreases as V increases



Thus, concave up curve for IV graph

- ✧ Filament lamps' R increases as V decreases



Thus, concave down curve for IV graph

18.4 Resistivity

Law of Resistance

- The resistance R of any wire is directly proportional to its length and inversely proportional to its cross-sectional area
 - $R = \rho \frac{L}{A}$
 - R : Resistance (Ω)
 - ρ : Resistivity ($\Omega \cdot \text{m}$)
 - L : Length (m)
 - A : Cross-sectional area (m^2)
- Resistivity: Constant of proportionality
 - Depends on the material used
 - Somewhat depends on purity, heat treatment, temperature, and other factors
 - Silver has lowest resistivity, but very expensive
 - Copper is close, and much less expensive, so used for most wires
 - Aluminum although higher resistivity, but less dense, so preferable for transmission lines

Temperature Dependence of Resistance & Resistivity

- For metal conductors, resistance increases with temperature
 - $T \uparrow$, atoms move more rapidly and are arranged in a less orderly fashion, so they interfere more with the flow of electrons, making them harder to move past
- Thus, if temperature change not too great, resistivity also increases nearly linearly with temperature
 - $\rho_T = \rho_0[1 + \alpha(T - T_0)]$
 - ρ_T : Resistivity at temperature T ($\Omega \cdot \text{m}$)

- ρ_0 : Resistivity at reference temperature T_0 (we usually use 20°C)
- α : Temperature Coefficient ($1/^\circ\text{C}$)
- However, for semiconductors, resistance and resistivity decreases with temperature increase
 - $T \uparrow$, some electrons that were normally not free become free and can contribute to the current flow
 - Thus, the temperature coefficient for semiconductors can be negative
- The resistance can also be applied with a similar formula (just by multiplying $\frac{L}{A}$ on both sides)
 - $R_T = R_0[1 + \alpha(T - T_0)]$

18.5 Electric Power

Energy Transformation

- Electrical Devices: Convert EE into other forms of energy (mechanical, thermal etc.)
- In an ordinary lightbulb, only a few percent of the energy is transformed into visible light, and rest of the >90% into thermal energy
 - Many collisions occur between moving electrons and atoms
 - In easy collision, part of the electron's KE is transferred to the atom
 - The KE of the atoms increases and hence the temperature of wire increases
 - This increased TE can be transferred as heat by conduction, convection, radiation, or as light

Power

- Power: The rate of energy transformed
 - $P = \frac{E}{t} = \frac{QV}{t} = IV$
 - $P = IV = I(IR) = I^2R$
 - $P = IV = \left(\frac{V}{R}\right)V = \frac{V^2}{R}$
 - Unit: Watt (W); $1\text{ W} = 1\text{ J/s}$; $1\text{ kW} = 1000\text{ W}$; $1\text{ hp} = 746\text{ W}$
- Voltage Rating: The designed voltage
- Power Rating: The power when supplied with the designed voltage
- Power dissipated: Actual power
 - The greater the actual power used by a lightbulb, the brighter it is

Energy

- $E = Pt$
 - Unit: J
 - $1\text{ kWh} = (1000\text{ W})(3600\text{ s}) = 3.6 \times 10^6\text{ J}$

18.6 Power in Household Circuits

Fuses and Circuit Breakers

- Overloaded: When a wire carries more current than is safe
- Fuse: A length of wire designed to melt and so breaking a circuit when the current passing through it goes above a certain level
 - Simple and cheap; material should have low melting point and high conductivity
 - The thicker the fuse wire the greater is the current required to cause it to melt
 - Rating current: Labelled maximum effective current designed to allow
 - ✧ $I_{effective} > I_{rating}$, then melt and burn
 - When a fuse burns out, it cannot be used again
- Circuit Breaker: A safety device that uses an electromagnet/bimetallic strip to shut off the circuit when the current goes above a certain value
 - Act more quickly than fuses
 - Bimetallic Strip Circuit Breaker: Heating of strip causes the strip to bend so far to one side that the notch in the spring-loaded metal strip drops down over the end
 - Electromagnet Circuit Breaker: Too much current causes the electromagnet to produce a magnetic field strong enough to open the switch
 - Can be reset using the outside switch
- Two possibilities if a circuit repeatedly burns out a fuse or opens a circuit breaker
 - Too many devices drawing current in that circuit
 - Short circuit (Resistance very small so current very large)
- Household circuits are parallel circuits
 - There are several paths for the current to take, and the voltage across each component is the same
 - Advantage:
 - ✧ If one light goes out, others burn brightly without change
 - ✧ As lights are added, current flowing through each light does not change
 - Disadvantage:
 - ✧ More difficult to design and build

18.7 Alternating Current

DC and AC

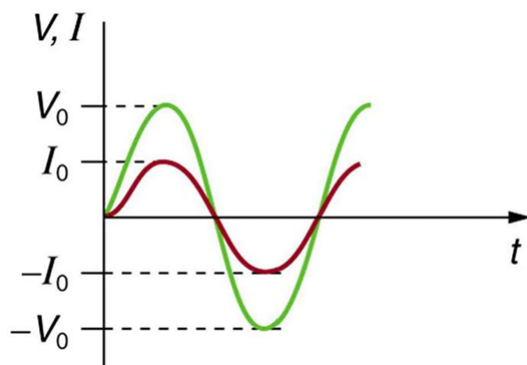
- Direct Current (dc): The current moves steadily in one direction
 - First found by **Thomas Alva Edison**
 - E.g. Battery
- Alternating Current (ac): Reverses direction many times per second and is commonly sinusoidal
 - First found by **Nikola Tesla**
 - E.g. Electric generators

Voltage for ac

- $V = V_0 \sin 2\pi ft = V_0 \sin \omega t$
 - V : Potential V oscillates between $+V_0$ and $-V_0$ (V)
 - V_0 : Peak voltage (V)
 - f : Frequency; number of complete oscillations made per second (1/s)
 - ω : Angular frequency ($=2\pi f$) (rad/s)

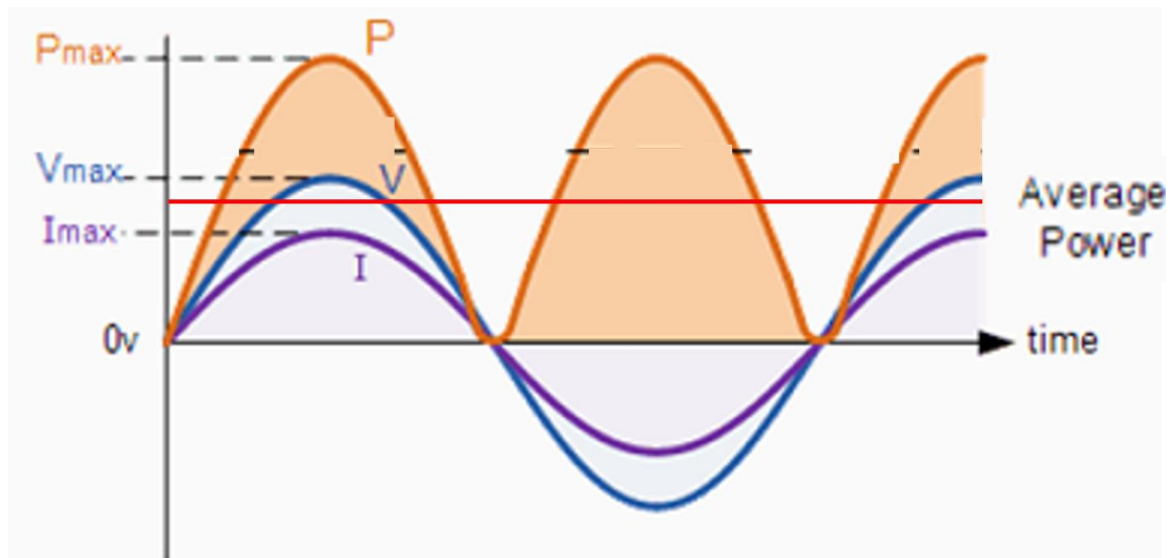
Current for ac

- $I = \frac{V}{R} = \frac{V_0}{R} \sin \omega t = I_0 \sin \omega t$
 - I_0 : Peak current (A)
- Current is considered positive when electrons flow in one direction and negative when they flow in the opposite direction
- Average current is 0



Power for ac

- $P = I^2 R = I_0^2 R \sin^2 \omega t$
- Power is always positive since current is squared
- $\bar{P} = \frac{1}{2} I_0^2 R = \frac{1}{2} \frac{V_0^2}{R}$
 - $\bar{P}$: Average power (W)
- $\bar{I}^2 = \frac{1}{2} I_0^2$
 - $I_{rms} = \sqrt{\bar{I}^2} = \frac{I_0}{\sqrt{2}} \approx 0.707 I_0$
- $\bar{V}^2 = \frac{1}{2} V_0^2$
 - $V_{rms} = \sqrt{\bar{V}^2} = \frac{V_0}{\sqrt{2}} \approx 0.707 V_0$
- rms (root-mean-square) values are sometimes called the effective values
 - $\bar{P} = I_{rms} V_{rms}$
 - $\bar{P} = \frac{1}{2} I_0^2 R = I_{rms}^2 R$
 - $\bar{P} = \frac{1}{2} \frac{V_0^2}{R} = \frac{V_{rms}^2}{R}$
- Thus, a direct current whose values of I and V equal the rms values of I and V for an alternating current will produce the same power

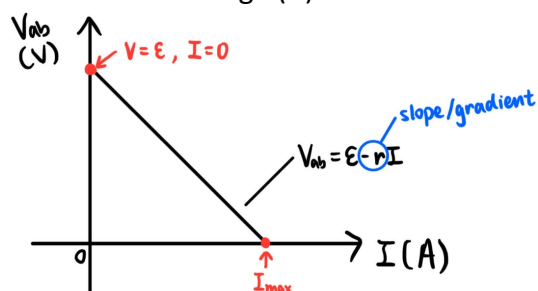


- Period of power is half of period of voltage and current

Chapter 19 – DC Circuit

19.1 EMF and Terminal Voltage

- Electromotive Force (emf): The potential difference between the terminals of such a source when no current flows to an external circuit
 - The source of emf transforms one type of energy (chemical, mechanical, light etc.) into EE
 - E.g. Battery, electric generator
- Internal resistance: The resistance of the battery itself
 - Usually small, but it increases as the battery ages
- Terminal Voltage: The voltage measured between two points a and b (Actual voltage)
 - $V_{ab} = V_a - V_b$
- When no current is drawn from battery:
 - $V_{ab} = \varepsilon$
- When a current flows naturally from the battery, there is an internal drop in voltage:
 - $V_{ab} = \varepsilon - Ir$
 - V_{ab} : Terminal voltage (V)
 - ε : emf (V)
 - Ir : Internal Voltage (V)

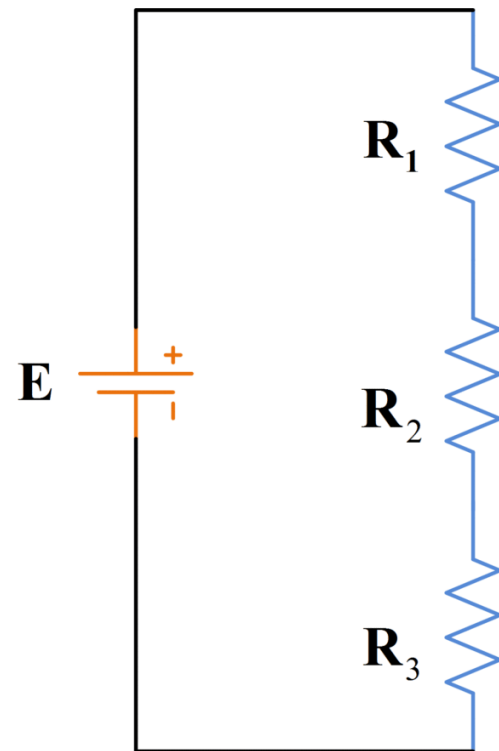


- $\therefore V_{ab} = IR$
 - $\therefore IR = \varepsilon - Ir$
 - $\varepsilon = I(R + r)$
 - ✧ A better exemplification of Ohm's Law
- $P_T = P_R + P_r = I^2 R + I^2 r = I^2 (R + r)$
- $P_T = P_R + P_r = \frac{V_{ab}^2}{R} + \frac{V_r^2}{r}$

19.2 Resistors in Series and in Parallel

Series Circuit

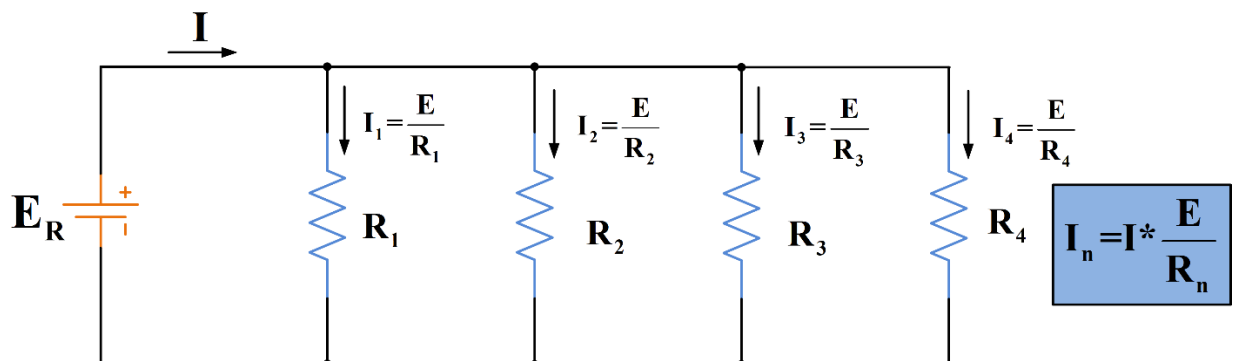
- Connected in series: When two or more resistors are connected end to end along a single path
- If one device is disconnected, current is stopped to all the others
- Same current passes through each resistor
 - $I = I_1 = I_2 = I_3$
 - According to conservation of charge, any charge that passes through R_1 will also pass through other resistors, $\therefore Q$ same, I same
- Total voltage is equal to the sum of the voltages across each resistor
 - $V = V_1 + V_2 + V_3$
 - According to energy conservation
- According to Ohm's Law:
 - $V = IR_1 + IR_2 + IR_3 = I(R_1 + R_2 + R_3) = IR_{eq}$
 - $R_{eq} = R_1 + R_2 + R_3$
 - R_{eq} : Equivalent Resistance: a single resistor with the resistance that is the same as the total resistance of the circuit; sum of separate resistors
 - $V_1 : V_2 : V_3 = R_1 : R_2 : R_3$
- Total power is equal to the sum of the power of individual resistors
 - $P_T = I^2 R_T = I^2 (R_1 + R_2 + R_3) = P_1 + P_2 + P_3$
 - $P_1 : P_2 : P_3 = R_1 : R_2 : R_3$ (Larger resistance, brighter lightbulb)
- Advantages:
 - Simple to design and build
- Disadvantages:
 - If one light goes out, all the lights go out
 - Lightbulbs become dimmer as more bulbs are added



$$R = R_1 + R_2 + R_3$$

Parallel Circuit

- Connected in parallel: The current from the source splits into separate branches or paths
- Same voltage across each resistor
 - $V = V_1 = V_2 = V_3$
- Total current is equal to the sum of the currents flowing through each resistor
 - $I = I_1 + I_2 + I_3$
- According to Ohm's Law:
 - $I = \frac{V^2}{R_1} + \frac{V^2}{R_2} + \frac{V^2}{R_3} = \frac{V^2}{R_1 + R_2 + R_3} = \frac{V^2}{R_{eq}}$
 - $R_{eq} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$
 - $I_1 : I_2 : I_3 = \frac{1}{R_1} : \frac{1}{R_2} : \frac{1}{R_3}$
- Total power is equal to the sum of the power of individual resistors
 - $P_T = I_T V = (I_1 + I_2 + I_3) V = P_1 + P_2 + P_3$
 - $P_1 : P_2 : P_3 = \frac{1}{R_1} : \frac{1}{R_2} : \frac{1}{R_3}$
- Advantages:
 - If there is a break in one branch, current can still move through other branches
 - Brightness of original lightbulbs does not change when more are added (ignore r)
- Disadvantages:
 - More difficult to design and build
 - Adding too much appliances will increase the current too much, which might heat up the wire and cause fire

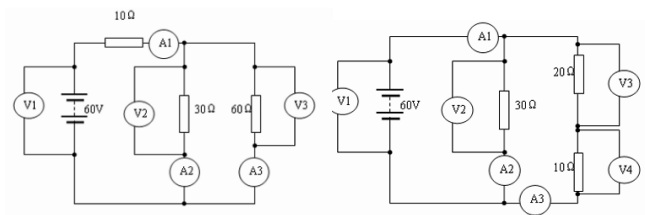


Chapter 19-DC Circuits

19.2(2) combined circuit

-Definition: A circuit which includes series and parallel branches.

-Examples of a combined circuit: left: Parallel-series right: Series-parallel

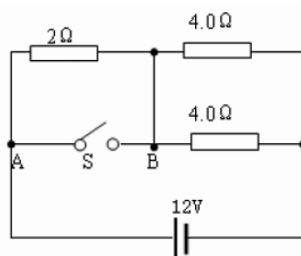


-Problem-Solving Strategies

1. Figure out the series or parallel relationships between the resistors in the circuit
2. Get R_{eq}
3. Get I_{total} 、 V_{total}
4. Find individual V, I and P

e.g.

- In the following circuit, first figure out the series or parallel relationships between the resistors when the switch is on or off.
- When the switch is off, calculate R_{eq} of the two 4.0Ω resistors, then calculate R_{eq} of the total .
- $I_{total} = \frac{V_{total}}{R_{eq}}$
- I of the 2Ω resistor = I_{total} ,V of the 2Ω resistor = $\frac{V_{total} \times 2\Omega}{R_{total}}$;the sum of I of the 4.0Ω resistors = I_{total} ,V of the 4.0Ω resistors are the same, which = $\frac{V_{total} \times R_{eq}}{R_{total}}$



Chapter 20-Magnetism

History of Magnetism

- Ancient Greeks and Chinese discovered: lodestone could attract iron objects and other lodestone.
- Chinese sailors employed magnets as navigation compasses approximately 900 years ago.

20.1 Magnets and Magnetic Fields

Magnets and Magnetic poles

-Magnet: An object surrounded by a magnetic field and has the property of attracting ferromagnetic materials (iron, nickel, cobalt).

- Two typical types of magnets are bar magnets and horseshoe magnets.
- A compass needle is simply a bar magnet which is supported at its center of gravity so that it can rotate freely.
- A magnet can line up in a north and south direction.

-Magnetic poles: the area of a magnet where the magnetic effect is strongest

-Every magnet is **polarized**. Every magnet has a north pole (N) and a south pole (S). N pole is the north-seeking pole, and S pole is the south-seeking pole.

-Magnetic force: interaction between magnetic poles.

- Between magnets, Like Poles Repel; Unlike Poles Attract.
- The closer the poles; the greater the force.

- Magnetic poles always appear in pairs. Monopoles of magnet do not exist.

- ✧ If a bar magnet is cut in half, you do not obtain isolated north and south poles. Instead, two new magnets are produced.

Magnetism and Magnetic domain

-Ferromagnetism: Iron, cobalt, nickel, gadolinium, and some of their oxides and alloys, show strong magnetic effects.

-Source of ferromagnetism:

- The spinning and orbiting motion of the electrons make each atom a tiny magnet. So each atom has a magnetic field.

- A cluster of billions of atoms that all have magnetic fields that are lines up in the same way. So they form a magnetic domain. Each domain is like a tiny magnet.
- Magnetization:
 - ✧ An unmagnetized piece of iron is made up of domains that are randomly arranged.
 - ✧ In a magnet, the domains are aligned in one direction (down), and may be altered in size by the magnetization process.

-Induced Magnetism:

- Both N/S magnetic poles attract metal such as iron, steel.
- Unmagnetized materials such as steel and iron acquire two opposite magnetic poles when placed in neighborhood of a magnet. That is how magnetism is induced.

-Making a magnet

- Place the unmagnetized material near a magnet. This gains weak magnetism.
- Stroke the materials with a magnet in one direction.
- Electromagnet: Make direct current passing through a coil of wire placed outside the materials.

- Hard and Soft magnetic materials: Depending on how well they keep their magnetism

- Hard magnetic materials
 - ✧ Difficult to magnetized but do not readily lose their magnetism.
 - ✧ Used for permanent magnets Examples: Steel, and alloys called Alnico, Alcomax and Magnadur
- Soft magnetic materials
 - ✧ Relatively easy to magnetize, but their magnetism is only temporary.
 - ✧ Used for temporary magnets. They are used in the cores of electromagnets and transformers, because their magnetic effect can be “switched” on or off or reversed easily. Examples: Iron(soft iron) and Mumetal

-Non-magnetic materials: Include metals such as brass, copper, zinc, tin, and aluminum, as well as non-metals.

-Demagnetize a magnet

- ✧ Drop a magnet on the floor or strike it with a hammer. The principle is to jar the domains into randomness and the magnet loses some or all of its magnetism.
- ✧ Heating a permanent magnet. The principle is that raising the temperature increases

the random thermal motion of atoms, which tends to randomize the domains. Above a certain temperature, it loses the property of ferromagnetism.

Magnetic field lines

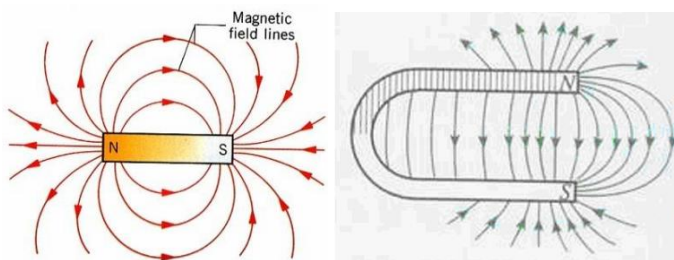
- Magnetic Field: the region around a magnet where the magnetic force is exerted.

-Experiments:

- ✧ Magnetic Field can be investigated using a small compass. The direction of the magnetic field B at any point is the same as the direction indicated by this compass.
- ✧ Magnetic Field can be mapped out by iron filings. The iron filings line up. Field B is strong where lines are dense and weak where lines are sparse.

- Magnetic field lines (They are three-dimensional): Imaginary lines to map out the magnetic field around a magnet. The field direction, is defined as the direction in which the force on a N pole would act.

Left : a bar magnet Right : a horseshoe magnet



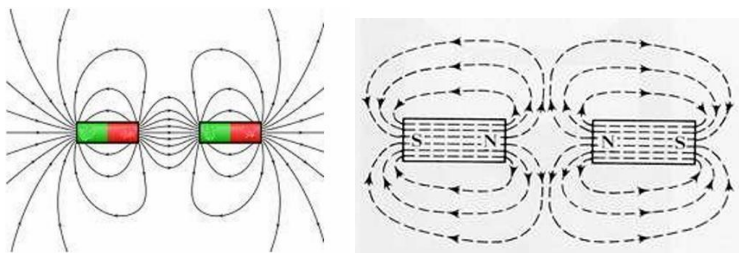
■ Direction of arrows

- ✧ Outside the magnet: from N to S
- ✧ Inside the magnet: from S to N
- ✧ A compass placed at any point: tangent to the line of force passing through that point

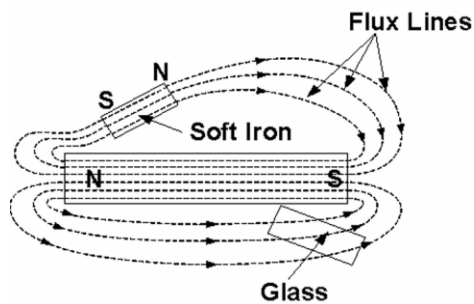
■ Density of lines represents how strong the magnetic field is.

■ The lines form complete loops and they never cross.

-Around two magnets (Combined field)



- Close to other materials



■ Glass: no reaction

■ Ferromagnetic material:

- ✧ The field lines become concentrated within the material.
- ✧ The end closest to the N pole of magnet becomes the S pole of the material.
- ✧ The end closest to the S pole of the magnet becomes the N pole of the material.
- ✧ The material is attracted to the magnet.

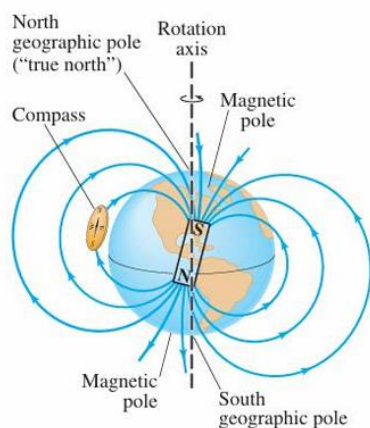
- Magnetic field lines (in 3D vision): Uniform magnetic field. Magnetic field between two wide poles of a magnet is nearly uniform, except near the edges.

Earth as a magnet

- Earth's magnetism is due to the circulation of the molten metal within Earth's core

- The earth is a large weak magnet with the magnetic S-seeking pole under a point in Canada.
- Earth's magnetic poles are not the same as the geographic poles (true south/north). They are 1200km apart.

- Magnetic Declination: The angular difference between the direction of a compass needle (which points along the magnetic field lines) at any location and true (geographical) north. Magnetic declination varies with location.



- The angle of dip or The angle of inclination: The angle that the Earth's magnetic field makes

with the horizontal at any point. The Earth's magnetic field at most locations is not tangent to the Earth's surface.

- Magnetic poles reverse themselves periodically. Earth's north magnetic pole has been moving over time, on the order of 10 km per year in recent decades. Magnetism in rocks solidified at various times in the past (age determined by radioactive dating) suggested that poles reversed direction 400 times over the last 330 million years.

20.2 Electric currents produce magnetic fields

Magnetic field around a wire

- Oersted's Experiment: Oersted(1777-1851) discovered in 1820. This experiment suggests that magnetism and electricity are related.

- The compass needle deflects if (and only if) the wire carries an electric current
- Whenever he turned on the electricity, the compass needles lined up in a circle around the wire.
- Iron filings also align along the direction of the magnetic field lines near a straight current carrying wire.

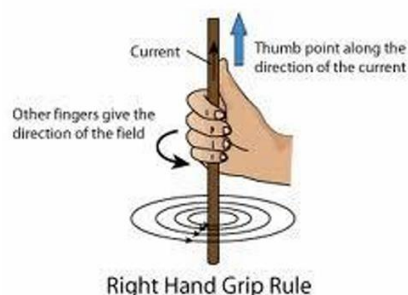
- Magnetic field around a wire: If an electric current is passed through a wire, a weak magnetic field is produced.

The magnetic field has these features:

- the magnetic field lines are circles
- the field is strongest close to the wire
- increasing the current increases the strength of the field.

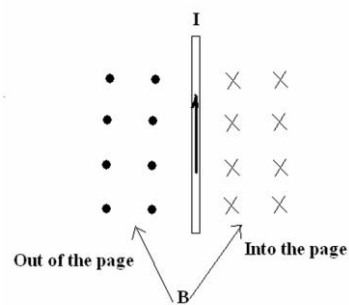
- Direction of the current (I) determines the direction of the magnetic field (B). Use 1st Right-Hand Rule for field direction.

- Grip the wire with your right hand.
- Thumb points in the conventional current direction (I).
- Curved fingers: in the direction of the magnetic field (B).



■

- Important symbols:



Magnetic field due to a long straight wire (section 20.5)

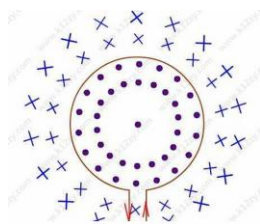
$$B = \frac{\mu_0 I}{2\pi r}$$

μ_0 = permeability of free space = $4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$

Standard unit of magnetic field B: Tesla(T)

- Combine B: Find out the magnitude and direction of B when 2 wires exist.

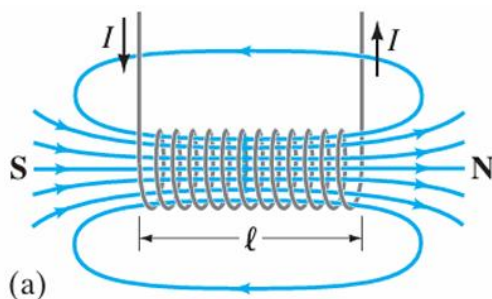
- Coiled Wire: Magnetic Field of a current-carrying loop.



Magnetic field for coils (section 20.7 solenoids)

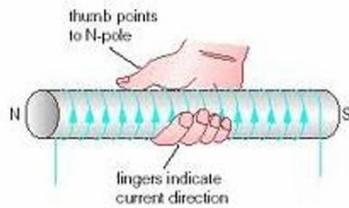
- Solenoid: A current-carrying coil of wire with many loops

- Magnetic field lines bunched up inside the loop
- Increase the number of the loops $\Rightarrow$ Increase the field



A solenoid acts like a magnet; one end can be considered the north pole and the other the south pole. Increasing the current increases the strength of the field. The magnetic field is temporary (turned on and off by switching the current on and off). The north and south poles change with the direction of the current

- find poles of the solenoid: 2nd Right-hand rule. You can also use it to find the direction of current in the wires.



- ✧ Grip the coil with your right hand
- ✧ Your fingers point in the conventional current direction
- ✧ Your thumb points towards the N pole of the coil

- Electromagnet: If a piece of iron is placed inside a solenoid, the magnetic field is increased greatly because the iron becomes a magnet. The resulting magnetic field is the sum of the field due to the current and the field due to the iron, and can be hundreds or thousands of times the field due to the current alone.

- Properties of electromagnets: *basically similar to solenoids except the last one.*

- ✧ can be switched on/off freely.
- ✧ I reverses, B reverses.
- ✧ I increases, B increases
- ✧ No. of loops increases, B increases
- ✧ soft iron core added, B increases

- Applications of electromagnets: Electromagnetic Crane, Magnetic Circuit Breakers, Doorbell, the magnetization and demagnetization of magnets.

- Magnetize a magnet: by inserting the material into the solenoid
- Demagnetize a magnet: Slowly pull a magnet out of a solenoid through which an alternating current is passing. AC produces a magnetic field which changes direction very rapidly and throws the atoms in the magnet out of line

20.3 Force on an electric current in a magnetic field

Force on a current in a Magnetic Field

- Ampere's Force: A magnet exerts a force on a current-carrying wire.

- Formula: $F = I l \perp B = I l B \sin \theta$

θ : angle between the current direction and the magnetic field

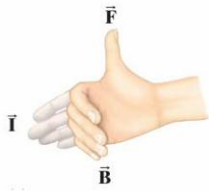
$F_{\max} = I l B$ when $\theta = 90^\circ$ (wire $\perp$ magnetic field)

$F_{\min} = 0$ when $\theta = 0^\circ$ (wire $\parallel$ magnetic field)

standard unit of B: tesla(T), $1\text{T} = 1\text{N/A}\cdot\text{m}$ (derived from this formula)

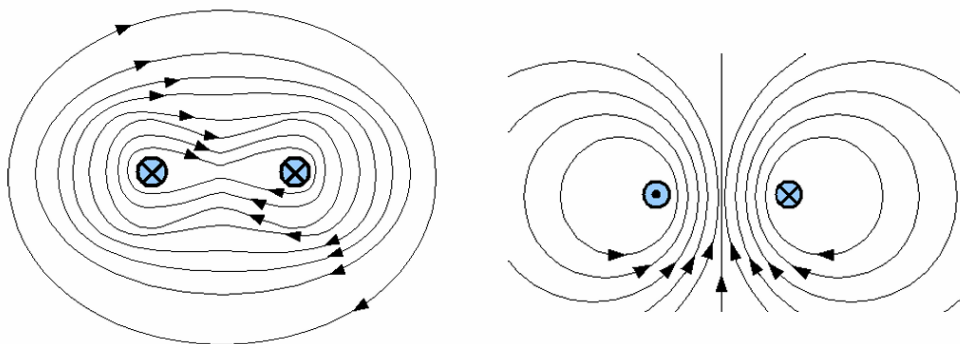
Earth's magnetic field: $5 \cdot 10^{-5} \text{ T} = 0.5 \text{ gauss}$ ($1 \text{ gauss} = 1\text{G} = 1 \cdot 10^{-4}\text{T}$). The strongest permanent magnets have fields near 2 T; superconducting electromagnets may attain 10 T or more.

-Find the direction of the force (or the current/field): The 3rd right hand rule (RHR)



Force between two parallel wires (section 20.6)

-the magnetic field:



- The Force between two parallel wires:

■ Direction of the force:

- ✧ current in the same direction -attraction
- ✧ current in opposite direction- repulsion

■ Magnitude of the force: (derived from Ampere's law)

$$B_1 = \frac{\mu_0 I_1}{2\pi d} \quad F_{1 \text{ on } 2} = I_2 l B_1 = \frac{\mu_0 I_1 I_2}{2\pi d} l$$

- Applications include: Galvanometer, Motor and loudspeaker

20.4 Force on an electric charge moving in a magnetic field

Force on Electric Charge Moving in a Magnetic Field

- Lorentz force: The force that free moving charged particles experience when they pass through a magnetic field.

- Magnitude of the force: $F = qvB\sin\theta$

θ is the angle between the directions of v and B

Derivation: If N such particles of charge q pass by a given point in time, they constitute a current $I = Q/t = Nq/t$

$$l/t = v$$

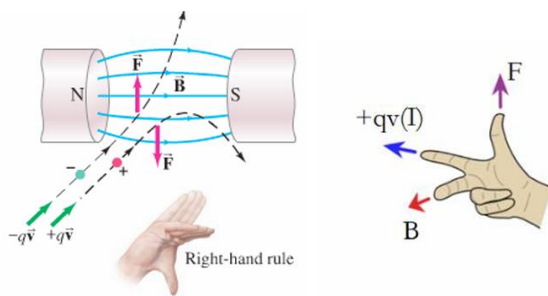
$$F = IlB\sin\theta = \frac{NqlB\sin\theta}{t} \rightarrow F = qvB\sin\theta \bullet$$

$F_{\max} = qvB$ when $\theta = 90^\circ$ (charge moves $\perp$ magnetic field)

$F_{\max} = 0$ when $\theta = 0^\circ$ (charge moves $\parallel$ magnetic field)

$$B = \frac{F}{qv} = \frac{F}{il} \rightarrow 1 \text{ T} = \frac{1 \text{ N}}{\text{C} \cdot \text{m/s}} = \frac{1 \text{ N}}{\text{A} \cdot \text{m}}$$

- Direction of the force: Right Hand Rule 3 (almost the same as the one to find the direction of Ampere force)



The path of a charged particle moving in a plane perpendicular to a uniform magnetic field

- If a charged particle is moving perpendicular to a uniform magnetic field, its path will be a circle. Because force is always perpendicular to motion.

$$F_{\text{net}} = F_C = F_B$$

$$\frac{mv^2}{r} = qvB$$

$$r = \frac{mv}{qB} \rightarrow r \propto v$$

$$T = \frac{2\pi r}{v} = \frac{2\pi m}{qB}, \text{ T is not related with v.}$$

- cyclotron frequency of a particle in a field: this is the frequency at which particles revolve in a cyclotron.

$$f = \frac{1}{T} = \frac{qB}{2\pi m}$$

- Aurora: Charged ions approach the Earth from the Sun (the "solar wind") and enter the

atmosphere mainly near the poles, ionizing the air, and as the electrons recombine with atoms, so light is emitted